Construction Chemicals

SUBMITTAL DOCUMENT

O MIX LAM HD Pro

Polyurethane Waterproofing
System for Trafficked Areas



Elastic
350 % elongation

Waterproof
PU membrane

Solvent-Free
PU & epoxy

3 Layers
Primer – filler – PU

Project	
Main Contractor	
Consultant	
Submittal No. / Rev.	00
Date	Nov 2026

Construction Chemicals

Contents

1. System Method Statement 3

2. TDS — O-Prime SF 11

3. TDS — O-Putty 14

4. TDS — Roofguard PU 17

5. Test Reports 21

6. Approvals & Certificates 22

METHOD STATEMENT

O MIX LAM HD Pro — Polyurethane Waterproofing System for Trafficked Areas

Project Name		**Document No.**	OMX-SMS-12
Main Contractor		**Revision**	01
Consultant		**Date of Issue**	Nov 2026
System	O MIX LAM HD Pro	**Status**	For Approval

1. Purpose

This method statement defines the sequence, resources, inspection and test plan and safety measures for the installation of the O MIX LAM HD Pro system on site. It describes how each layer of the system is prepared, mixed, applied, inspected and protected so that the completed system achieves the performance stated in the O MIX Technical Data Sheets of its components.

2. Scope of Work

O MIX LAM HD Pro is a polyurethane waterproofing system for industrial floors, bridge decks, ramps, wet areas and podium decks. A solvent-free epoxy primer and epoxy filling layer are followed by a solvent-free, elastic polyurethane membrane.

- Substrate inspection, testing and acceptance.
- Surface preparation and repair of defects.
- Stage 1: Primer — O-Prime SF.
- Stage 2: Filling layer — O-Putty.
- Stage 3: PU membrane (2 coats) — Roofguard PU.
- Curing, protection, final testing and handover.

2.1 System build-up

Stage	Product	Function	Consumption
1	O-Prime SF	Primer	0.10 – 0.15 kg/m²
2	O-Putty	Filling layer	0.25 – 0.30 kg/m²
3	Roofguard PU	PU membrane (2 coats)	0.5 kg/m²

2.2 Typical areas of use

- Industrial floors and ramps.
- Bridge and podium decks.
- Wet areas.
- Roofs with foot traffic.

3. Definitions

Term	Meaning
System	All layers of the build-up applied in the stated sequence.
Hold point	Stage at which work stops until QA/QC (and the consultant, where required) approve.
Recoat window	Minimum and maximum time between two layers for proper inter-layer bonding.
Dew point	Temperature at which moisture condenses; the substrate must be at least 3 °C above it.
Mock-up / trial area	Area applied before the main works to approve materials, method and appearance.
Pot life	Time a mixed multi-component material remains workable.

4. Roles & Responsibilities

Role	Responsibility
Project Manager	Planning, resources, approvals and coordination with the consultant and main contractor.
Site Engineer	Daily supervision; confirms substrate readiness, climate, mixing, consumption and recoat windows; keeps daily records.
QA/QC Inspector	Carries out tests at each hold point, records results and signs the inspection checklist.
HSE Officer	Toolbox talks, PPE, ventilation, safe access and storage; stops unsafe work.
Applicator Foreman	Leads the application team and follows this method statement; reports deviations.
O MIX Technical Representative	Technical support, site visits and advice on product use when requested.

5. Materials

Product	Description	Packaging
O-Prime SF	Solvent-Free Epoxy Primer & Sealer for Concrete Floors	5 kg sets: Part A in pail + Part B in jerrycan.
O-Putty	Two-Component Epoxy Filler for Concrete, Steel & Equipment	5 kg sets (Part A + Part B), metal tins.
Roofguard PU	Solvent-Free Two-Component Polyurethane Waterproofing Membrane	20 kg sets (10 kg A + 10 kg B) and 5 kg sets (2.5 kg + 2.5 kg).

5.1 Key product data

Product	Mixing ratio	Pot life / working time	Recoat / drying
O-Prime SF	As recommended — pre-weighed sets	40 min	10 hours (max. 24 hours)
O-Putty	1 : 1 by volume	60 min	24 hours
Roofguard PU	1 : 1 by weight — pre-weighed sets, mix as recommended	40 min	16 – 48 hours

6. Tools & Equipment

- Diamond grinder and/or shot blaster with dust extraction; industrial vacuum
- Slow-speed drill mixer (300 – 500 rpm) with spiral paddle
- Rollers, brushes, squeegees, notched trowels and steel trowels as required by each stage
- Scales and measuring containers for part mixing (where permitted)
- Thermo-hygrometer, surface thermometer and dew-point calculator
- Moisture meter and plastic sheets for moisture testing
- Masking tape, protective sheeting, barriers and warning signs
- Lighting and extension cables with residual current protection
- Scissors and knives for reinforcement
- Hose and plugs for flood testing

6.1 Personal protective equipment

- Safety shoes, helmet and high-visibility vest
- Chemical-resistant gloves (nitrile) and safety goggles
- Coveralls with long sleeves
- Dust masks (P2) during grinding and powder mixing
- Respirators with organic vapour cartridges for solvent-containing products and confined spaces
- Knee pads for floor work

7. Storage & Handling

- Store all products in original, unopened packaging in a dry, shaded and ventilated store at the temperatures stated in each TDS.
- Keep Part A, Part B (and Part C) of each kit together; do not mix components from different batches.
- Protect liquids from frost and direct sun, and powders from moisture.
- Store solvent-containing products away from heat, sparks and ignition sources.
- Use materials on a first-in, first-out basis and record batch numbers for each area.
- Condition materials to +15 °C to +25 °C for 24 hours before use in hot or cold weather.

8. Pre-Start Requirements

- Approved material submittal, this method statement and the inspection and test plan.
- Approved drawings showing areas, joints, upstands, drains and details.
- Work area handed over clean, free of other trades, with safe access, lighting and power.
- Mock-up / trial area executed and approved for system, colour and finish.
- Toolbox talk held with the team on this method statement and HSE measures.
- Climate monitoring equipment calibrated and available.

9. Substrate Inspection & Acceptance (Hold Point)

Check	Acceptance criteria	Method
Concrete age	≥ 28 days (cement-based substrates)	Site records
Compressive strength	≥ 25 N/mm²	Rebound hammer (ASTM C805) or cores
Surface tensile (pull-off) strength	≥ 1.5 N/mm²	Pull-off test ASTM D7234
Moisture content	< 4 % (no moisture under plastic sheet after 16 h)	Moisture meter; plastic sheet test ASTM D4263
Falls to drains	No ponding; minimum falls as per design	Visual / water test
Upstands and details	Coves formed; penetrations sealed	Visual
Climate	Substrate ≥ 3 °C above dew point; RH ≤ 85 %	ISO 8502-4 / thermo-hygrometer

10. Surface Preparation

1. Remove laitance, curing compounds, oil, paint and loose material by shot blasting or diamond grinding to the required profile.
2. Open cracks to a V-groove and remove dust; repair holes, voids and spalls.
3. Grind high spots and remove sharp edges; form 45° coves at wall / floor junctions where required.
4. Vacuum all dust with an industrial vacuum immediately before priming.
5. Check falls and drains; repair ponding areas.
6. Reinforce joints and cracks with O-Flex PVC or reinforcement as detailed.
7. Mask and protect adjacent surfaces, drains and fixtures.
8. Record substrate test results and obtain QA/QC approval before priming (hold point).

11. Application Sequence

Apply the system stage by stage in the order below. Each stage must be inspected and approved before the next stage starts, and the next stage must be applied within the recoat window stated.

11.1 Stage 1 — Primer: O-Prime SF

Consumption: 0.10 – 0.15 kg/m². Time before next stage: Min. 10 h, max. 24 h.

1. Check batch numbers and condition the kit to +15 °C to +25 °C.
2. Pour all of Part B into Part A.
3. Mix 2 – 3 minutes below 300 rpm until uniform.
4. Mix complete sets only; do not thin.
5. Apply by roller at 0.10 – 0.15 kg/m² without puddles.
6. Apply a second coat on very porous substrates.
7. Overcoat with the next system layer within 24 hours, or broadcast 0.1 – 0.4 mm quartz sand into the wet primer.

11.2 Stage 2 — Filling layer: O-Putty

Consumption: 0.25 – 0.30 kg/m². Time before next stage: 24 h, then light sanding.

1. Check batch numbers and condition the kit to +15 °C to +25 °C.
2. Take equal volumes of Part A and Part B.
3. Mix on a clean board until uniform in colour, with no streaks.
4. Press the filler firmly into cracks and voids with a steel spatula.
5. Skim the surface at 0.25 – 0.30 kg/m² where used as a filling layer.
6. After 24 hours, sand lightly and remove dust before overcoating.

11.3 Stage 3 — PU membrane (2 coats): Roofguard PU

Consumption: 0.5 kg/m². Time before next stage: Second coat within 48 h; full cure 7 days.

1. Check batch numbers and condition the kit to +15 °C to +25 °C.
2. Stir Part A, add the complete Part B (1 : 1 by weight).
3. Mix 2 – 3 minutes until uniform.
4. Reinforce details with mesh in the first coat.
5. Apply the first coat by squeegee or notched trowel.
6. Apply the second coat within 48 hours.
7. Broadcast quartz sand on the last coat where specified.

12. Hold Points & Recoat Windows

Stage	Before starting	Waiting time before next stage	Approved by
1. O-Prime SF	Substrate accepted (Section 9)	Min. 10 h, max. 24 h	QA/QC
2. O-Putty	Previous stage inspected and approved	24 h, then light sanding	QA/QC
3. Roofguard PU	Previous stage inspected and approved	Second coat within 48 h; full cure 7 days	QA/QC

13. Curing, Protection & Return to Service

- Protect each layer from rain, dew, dust, water and traffic until it has cured.
- Barricade the area and display warning signs during application and curing.
- Keep ventilation running in enclosed areas until full cure.
- Allow full cure before flood testing and before installing protection screed, tiles or separation layers.

14. Quality Control & Inspection Test Plan

Activity	Inspection / Test	Frequency	Record
Material receipt	Product, batch, expiry, packaging condition	Each delivery	Material inspection report

Activity	Inspection / Test	Frequency	Record
Climate	Air / substrate temperature, RH, dew point (ISO 8502-4)	Start of shift and every 4 h	Daily climate log
Substrate	Strength, pull-off (ASTM D7234), moisture (ASTM D4263)	Each area (hold point)	Inspection request
Mixing	Complete kits, ratio, mixing time	Each mix	Daily record
Consumption	Material used vs area	Each stage	Daily record
Adhesion	Pull-off test on trial area / as specified (ASTM D7234 or D4541)	As specified	Test report
Watertightness	Flood test 48 h (ASTM D5957)	Each roof / area	Flood test report
Final inspection	Appearance, continuity, finish, details	Each area	Handover checklist

15. Defects & Remedies

Defect	Likely cause	Remedy
Poor adhesion / delamination	Weak, dusty or damp substrate; recoat window exceeded	Remove, re-prepare and reapply; abrade and re-prime if window exceeded
Pinholes / bubbles	Outgassing; air trapped in mixing; thick application	Seal porous areas; mix slowly; spiked roller; sand and recoat
Blushing / tacky surface	High humidity or low temperature during cure	Wash, abrade and recoat in suitable conditions
Soft spots	Incorrect ratio or incomplete mixing	Remove and reapply with complete kits
Leakage in flood test	Defective details or thin areas	Locate, repair, reinforce and re-test
Colour / shade variation	Different batches; uneven application	Use one batch per area; apply additional coat

16. Health, Safety & Environment

Hazard	Control measures
Skin / eye contact with resins and cement	Gloves, goggles, coveralls; eye-wash station
Inhalation of dust and vapours	Dust extraction; ventilation; masks / respirators
Fire (solvent vapours)	No smoking or hot work; fire extinguisher at the work area
Slips and trips	Housekeeping; barricades and warning signs
Manual handling	Correct lifting; mechanical aids for drums and pails
Electrical equipment	Inspected tools; protected cables with RCD
Work at height	Edge protection, harness and anchorage

- Read the Safety Data Sheet of each product before use.
- Keep first-aid kit and eye-wash on site; follow the first-aid measures in each TDS.

16.1 Waste & environment

- Collect empty containers and residues in designated bins.
- Allow small quantities of mixed resin to cure before disposal; dispose of according to local regulations.
- Do not discharge materials or washings into drains or soil.

17. Cleaning

- Clean tools of water-based and cementitious products with water immediately after use.
- Clean tools of epoxy and PU products with the recommended thinner before the material cures.
- Cured material can only be removed mechanically.

18. Limitations

- Do not apply outside the temperature and humidity limits stated in the TDS of each product.
- Do not apply on wet surfaces or when rain or dew is expected before the layer has cured.
- Do not mix or substitute products from other systems without written approval from O MIX.
- Any deviation from this method statement must be approved by the Site Engineer and the consultant.

19. Maintenance

- Inspect the roof twice a year and after storms.
- Keep drains clear.
- Repair damage promptly with the same system.

Appendix A — Inspection Checklist

Inspection Item	Result	Remarks
Approved submittal and method statement on site	☐ Yes ☐ No	
Mock-up approved	☐ Yes ☐ No	
Substrate tests passed (strength, pull-off, moisture)	☐ Yes ☐ No	
Surface preparation approved	☐ Yes ☐ No	
Stage 1 — O-Prime SF applied and approved	☐ Yes ☐ No	
Stage 2 — O-Putty applied and approved	☐ Yes ☐ No	
Stage 3 — Roofguard PU applied and approved	☐ Yes ☐ No	
Recoat windows respected	☐ Yes ☐ No	
Final tests passed	☐ Yes ☐ No	
Area protected and handed over	☐ Yes ☐ No	

Inspected by	Signature	Date

Appendix B — Daily Application Record

Date	Area / zone	Stage	Batch No.	Qty used	Air / substrate °C	RH %	Sign

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O-Prime SF Data Sheet

O MIX LAM HD Pro — Submittal Document

Construction Chemicals

Product Datasheet
Edition: 01 | Date: Nov 2026
Flooring

O-Prime SF

Solvent-Free Epoxy Primer
& Sealer for Concrete Floors



Product Description

O-Prime SF is a high-performance, two-component, solvent-free epoxy primer that penetrates and seals concrete surfaces. It is the base coat for O MIX epoxy and polyurethane floor systems and can also be applied on steel and aluminium.

Uses

- Primer for epoxy and polyurethane floor systems.
- Sealing porous concrete and screeds.
- Dust-proofing of concrete floors.
- Primer on steel and aluminium substrates.
- Base coat in O MIX floor systems ST, MD, HD and HD Pro.

Advantages

- Solvent-free, 100 % solids.
- Deep penetration, seals pores.
- Excellent adhesion: 3 N/mm².
- High mechanical strength.
- Flash point 140 °C.

Application

1. **Surface Preparation**
   Concrete ≥ 28 days, dry (< 4 %). Grind or shot blast and vacuum all dust.
2. **Mixing**
   Add all of Part B to Part A and mix 2 – 3 min below 300 rpm.
3. **Application**
   Roll 0.10 – 0.15 kg/m². Overcoat within 24 h or broadcast sand.

Consumption

0.10 – 0.15 kg/m² per coat
Two coats on very porous substrates
Depends on substrate porosity.

Packaging

5 kg sets: Part A in pail + Part B in jerrycan.

Storage & Shelf Life

12 months in original, unopened packaging. Store dry at +5 °C to +35 °C, away from direct sunlight.

Technical Data

Typical values at 25 °C

Property	Typical Value	Test Method
Mixing ratio (A : B)	As recommended — pre-weighed sets	—
Solids content	100 %	—
Specific gravity	1.06	ASTM D1475
Flash point	140 °C	—
Pot life (25 °C)	40 min	—
Recoat interval	10 hours (max. 24 hours)	—
Foot traffic	24 hours	—
Adhesion to concrete	3 N/mm² (concrete failure)	ASTM D4541
Compressive strength	70 N/mm²	ASTM D695
Flexural strength	50 N/mm²	ASTM D790
Tensile strength	40 N/mm²	ASTM D638

Consumption

Substrate	Coats	Consumption
Dense concrete	1	0.10 – 0.15 kg/m²
Porous concrete / screed	2	0.10 – 0.15 kg/m² per coat
Steel / aluminium	1	0.10 – 0.15 kg/m²

Consumption is theoretical and varies with substrate porosity, profile and application method.

Application Conditions

- Ambient and substrate temperature: +10 °C to +35 °C.
- Substrate at least 3 °C above dew point.
- Relative humidity: max. 85 %.
- Substrate moisture content: < 4 %.
- Concrete at least 28 days old, ≥ 25 N/mm², pull-off ≥ 1.5 N/mm².

Application Notes

- Mix with a slow-speed mixer below 300 rpm to avoid air entrapment.
- Apply by roller or brush; do not leave puddles.
- Apply two coats on very porous substrates.
- Overcoat within 24 hours; otherwise broadcast 0.1 – 0.4 mm quartz sand into the wet primer.
- Do not thin.

Health & Safety

- Wear nitrile gloves, safety goggles and overalls.
- Uncured epoxy may cause skin sensitisation; avoid skin and eye contact.
- Ensure good ventilation in enclosed areas.

First Aid & Cleaning

- Eyes: rinse with clean water for 15 minutes and seek medical advice.
- Skin: wash with soap and water.
- Clean tools with epoxy thinner before the material cures.

The information in this datasheet is based on laboratory tests and practical experience and is given in good faith. Users should verify product suitability for their specific application. O MIX reserves the right to update this information without prior notice.

Construction Chemicals

O-Putty Data Sheet

O MIX LAM HD Pro — Submittal Document

Construction Chemicals

Product Datasheet
Edition: 01 | Date: Nov 2026
Flooring



O-Putty

Two-Component Epoxy Filler
for Concrete, Steel & Equipment

Product Description

O-Putty is a two-component, solvent-based epoxy filler for high-performance repairs. It fills cracks, voids and surface defects in concrete, steel and equipment, dries quickly and resists heat, chemicals and bacteria. Any epoxy coating can be applied over it after light sanding.

Uses

- Filling cracks and voids in concrete floors.
- Industrial, food and pharmaceutical floors.
- Garages, bridges and ramps.
- Repair of tanks, drums, pipes and equipment.
- Filling layer in O MIX floor systems.

Advantages

- Fast drying.
- Resistant to heat, chemicals and bacteria.
- High compressive strength: 60 N/mm².
- Excellent adhesion to concrete and steel.
- Can be overcoated with any epoxy.

Application

1. **Surface Preparation**
   Clean, dry and sound substrate. Remove dust, oil, rust and loose particles.
2. **Mixing**
   Mix Part A and Part B 1 : 1 by volume until uniform in colour.
3. **Application**
   Fill and level with a spatula or trowel. Sand lightly before coating.

Consumption

0.25 – 0.30 kg/m²
As a skim / filling layer
Higher on rough or damaged surfaces.

Packaging

5 kg sets (Part A + Part B), metal tins.

Storage & Shelf Life

12 months in original, unopened packaging.
Store dry and cool, away from heat and ignition sources.

Technical Data

Typical values at 25 °C

Property	Typical Value	Test Method
Mixing ratio (A : B)	1 : 1 by volume	—
Colour	White grey	—
Pot life (25 °C)	60 min	—
Drying time	24 hours	—
Full cure	7 days	—
Compressive strength	60 N/mm²	ASTM D695
Flexural strength	30 N/mm²	ASTM D790
Shelf life	12 months	—

Consumption

Use	Consumption
Skim / filling layer in floor systems	0.25 – 0.30 kg/m²
Crack and void filling	According to volume of defect

Consumption is theoretical and varies with substrate porosity, profile and application method.

Application Conditions

- Ambient and substrate temperature: +10 °C to +35 °C.
- Substrate at least 3 °C above dew point.
- Relative humidity: max. 85 %.
- Substrate moisture content: < 4 %.

Application Notes

- Mix equal volumes of Part A and Part B on a clean board until uniform.
- Apply with a steel spatula or trowel; press well into cracks.
- Allow 24 hours drying, then sand lightly before overcoating.
- Keep away from flames — contains solvent.

Health & Safety

- Contains solvent: keep away from sparks, flames and smoking.
- Wear gloves, goggles and overalls; use a mask in enclosed areas.
- Ensure good ventilation.

First Aid & Cleaning

- Eyes: rinse with clean water for 15 minutes and seek medical advice.
- Skin: wash with soap and water.
- Clean tools with epoxy thinner before the material cures.

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Construction Chemicals

Roofguard PU Data Sheet

O MIX LAM HD Pro — Submittal Document

Construction Chemicals

Product Datasheet
Edition: 01 | Date: Nov 2026
Waterproofing

Roofguard PU

Solvent-Free Two-Component
Polyurethane Waterproofing Membrane



Product Description

Roofguard PU is a two-component, solvent-free, cold-applied polyurethane waterproofing membrane. It forms a seamless, elastic film with 350 % elongation for roofs, terraces, wet areas, pools, tunnels and car parks, and can be reinforced with fibre mesh.

Uses

- Roofs, terraces and balconies.
- Bathrooms, kitchens and wet areas.
- Swimming pools and planters.
- Tunnels, basements and car parks.
- Top coat in the O MIX LAM HD Pro system.

Advantages

- Elongation 350 %.
- Solvent-free, 99.99 % solids.
- Seamless, waterproof membrane.
- Abrasion resistant.
- Cold applied.

Application

1. **Surface Preparation**
   Clean, sound substrate; repair cracks, form coves; prime with O-Prime SF (O-Prime WS on damp surfaces).
2. **Mixing**
   Add Part B to Part A (1 : 1 by weight) and mix 2 – 3 min.
3. **Application**
   Apply by squeegee or notched trowel in 2 coats; second coat within 48 hours.

Consumption

0.2 – 0.3 kg/m²
at 250 µm (2 coats)
Reinforced system up to 1.5 kg/m² (3 coats); 0.5 kg/m² in LAM HD Pro.

Packaging

20 kg sets (10 kg A + 10 kg B) and 5 kg sets (2.5 kg + 2.5 kg).

Storage & Shelf Life

12 months in original, unopened packaging. Store dry at +5 °C to +35 °C, away from direct sunlight. Protect from moisture.

Technical Data

Typical values at 25 °C

Property	Typical Value	Test Method
Mixing ratio (A : B)	1 : 1 by weight — pre-weighed sets, mix as recommended	—
Colour	Grey, beige or on request	—
Solids content	99.99 %	—
Specific gravity	1.34	—
Pot life	40 min	—
Drying time	16 – 48 hours	—
Full cure	7 days	—
Hardness	Shore A 50	—
Tensile strength	6 – 7 N/mm²	—
Elongation	350 %	—
Adhesion	1.3 N/mm²	—
Water absorption	0.95 %	—
Abrasion	67.3 mg loss	—
DFT (reference)	406 µm	—

Consumption

System	Coats	Consumption
Standard coating, 250 µm	2	0.2 – 0.3 kg/m²
LAM HD Pro top coat	1 – 2	0.5 kg/m²
Reinforced membrane	3 + mesh	up to 1.5 kg/m²

Consumption is theoretical and varies with substrate porosity, profile and application method.

Application Conditions

- Ambient and substrate temperature: +10 °C to +35 °C.
- Substrate at least 3 °C above dew point.
- Relative humidity: max. 85 %.
- Substrate clean and sound; concrete at least 28 days old.

Application Notes

- Prime with O-Prime SF; use O-Prime WS on damp substrates.
- Apply 2 coats; second coat within 48 hours.
- Reinforce joints and cracks with mesh embedded in the first coat; use O-Flex PVC at moving joints.
- Broadcast quartz sand on the last coat for an anti-slip finish.

Health & Safety

- Wear chemical-resistant gloves, safety goggles and overalls.
- Uncured epoxy may cause skin sensitisation; avoid skin and eye contact.
- Ensure good ventilation in enclosed areas.

First Aid & Cleaning

- Eyes: rinse immediately with plenty of clean water for at least 15 minutes and seek medical advice.
- Skin: remove contaminated clothing and wash with soap and water; do not use solvents on skin.
- Inhalation: move to fresh air; seek medical advice if symptoms persist.
- Ingestion: do not induce vomiting; seek medical advice immediately.
- Clean tools with PU thinner before the material cures.

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Test Reports

O MIX LAM HD Pro — Submittal Document

To be attached: laboratory test reports supporting the values in the Technical Data Sheets.

Construction Chemicals

Approvals & Certificates

O MIX LAM HD Pro — Submittal Document

To be attached: approvals and certificates, where available.